Gravitational Wave Predictions of the $\Sigma = 2\ln Q$ Framework: Quantitative Estimates for Current and Near-Future Detectors

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(Dated: March 24, 2026)

We derive three specific, testable gravitational wave predictions from the $\Sigma = 2\ln Q$ framework developed in the companion papers. (1) The Khronon ghost condensation mechanism predicts *no detectable scalar gravitational wave mode* in any current or planned detector: the Jeans frequency $f_J \sim 10^{-19}$ Hz lies far below all detector bands, and the DBI-saturated scalar strain is bounded by $h_s < \alpha_{\rm DBI}\, h_t < 0.05\, h_t$. (2) The exponential metric $g_{00} = -e^{-r_{\rm s}/r}$, derived from Petz bound saturation, produces a cumulative phase shift of $\Delta\Phi \sim 10^4$–10^6 radians in extreme-mass-ratio inspirals (EMRIs) observable by LISA, with a minimum detectable fractional deviation from Schwarzschild of $\alpha_{\rm min} \sim 10^{-6}$. (3) The framework predicts holographic scaling ($\alpha_{\rm dec} = 0$) for gravitational wave phase diffusion, distinguishable from string foam ($\alpha_{\rm dec} = -1$) and causal set ($\alpha_{\rm dec} = +1$) predictions by the Einstein Telescope and Cosmic Explorer.

INTRODUCTION

The $\Sigma = 2\ln Q$ framework [? ? ?] identifies gravitational entropy production with the quantum relative entropy (QRE) and predicts the exponential metric as the Petz-saturated geometry. While Paper 2 established three strong-field predictions (shadow size, QNM frequency shift, and echoes), several gravitational wave signatures remain to be quantified. Here we provide explicit numerical predictions that can be confronted with data from current LIGO–Virgo–KAGRA observations, the upcoming LISA mission, and next-generation detectors (Einstein Telescope, Cosmic Explorer).

PREDICTION 1: NO SCALAR GRAVITATIONAL WAVE MODE

Khronon perturbation sound speed

In the Blanchet–Skordis Khronon theory [? ?], the scalar field ϕ sits in a ghost condensation background with kinetic function $K(Q)$, where $Q = |\partial\phi|$ is the inverse Khronon lapse. The perturbation sound speed is

$$c_{\rm s}^2 = \frac{K'(Q_0)}{K'(Q_0) + 2\,Q_0\,K''(Q_0)} \xrightarrow{K'(Q_0)=0} 0\,. \tag{1}$$

Ghost condensation ($K'(Q_0) = 0$) gives $c_{\rm s}^2 = 0$ exactly [?], which is radiatively stable with quantum corrections $\delta c_{\rm s}^2 \sim (\mu/M_{\rm Pl})^2 \sim 10^{-122}$ [?].

The DBI completion [?] introduces a scale-dependent correction:

$$c_{\rm s}^2(k,a) = \alpha_{\rm DBI}\,\frac{(k/k_J)^2}{1+(k/k_J)^2}\,, \tag{2}$$

where $\alpha_{\rm DBI} = Q_0/(2\lambda_D^2)$ and the Jeans wavenumber is

$$k_J(a) = \sqrt{\tfrac{3}{2}\,\Omega_K\,H_0^2/a}\,, \tag{3}$$

with $\Omega_K \approx 0.265$ the Khronon energy fraction.

Jeans frequency

Evaluating at $a = 1$ (today) with $H_0 = 67.36$ km/s/Mpc:

$$k_J = 1.42\times10^{-4}\ {\rm Mpc}^{-1} = 4.59\times10^{-27}\ {\rm m}^{-1}\,. \tag{4}$$

The corresponding frequency is

$$\boxed{f_J = \frac{c\,k_J}{2\pi} = 2.2\times10^{-19}\ {\rm Hz}}\,, \tag{5}$$

which is 10^{10} times below nanohertz pulsar timing arrays and 10^{17} times below the LIGO band.

At all detector frequencies ($f \gg f_J$), the ratio $k/k_J \gg 1$ and $c_{\rm s}^2 \to \alpha_{\rm DBI}$. The CMB constraint from Paper 3 gives $\alpha_{\rm DBI} < 0.05$; the ghost condensation natural value is $\alpha_{\rm DBI} \lesssim 0.01$.

Scalar mode amplitude

The scalar gravitational wave strain couples to the detector through the breathing mode polarization. In the Blanchet–Skordis theory, the scalar-to-tensor amplitude ratio is bounded by [? ?]

$$\frac{h_{\rm scalar}}{h_{\rm tensor}} \lesssim \alpha_{\rm DBI} < 0.05\,. \tag{6}$$

At Advanced LIGO sensitivity $h \sim 10^{-23}$, the scalar mode is $h_s \lesssim 5\times10^{-25}$, well below the noise floor.

Prediction.—No scalar gravitational wave mode is detectable by any current or planned detector (LIGO, LISA, ET, CE, PTA). The detection of a propagating scalar mode at any frequency would falsify the ghost condensation mechanism $K'(Q_0) = 0$ that underpins the framework.

TABLE I. Comparison of the exponential and Schwarzschild metrics at the same areal radius $R = u\,GM/c^2$. The fractional orbital frequency difference $\delta\omega/\omega$ is computed from the effective potential.

u	$g_{00}^{\rm Schw}$	$g_{00}^{\rm exp}$	$\delta\omega/\omega$
6	-0.6667	-0.6643	$+1.22\%$
8	-0.7500	-0.7491	$+0.58\%$
10	-0.8000	-0.7996	$+0.34\%$
15	-0.8667	-0.8666	$+0.13\%$
20	-0.9000	-0.9000	$+0.07\%$
50	-0.9600	-0.9600	$+0.001\%$

PREDICTION 2: EMRI PHASE SHIFT FROM EXPONENTIAL METRIC

Metric comparison at the same areal radius

The exponential metric in isotropic coordinates [**?**],

$$ds^2 = -e^{-r_{\rm s}/r_{\rm iso}}\,c^2\,dt^2 + e^{+r_{\rm s}/r_{\rm iso}}\big(dr_{\rm iso}^2 + r_{\rm iso}^2\,d\Omega^2\big)\,, \quad (7)$$

must be compared with Schwarzschild at the same circumferential (areal) radius R. The coordinate transformation $R = r_{\rm iso}\,\exp\big(GM/(c^2 r_{\rm iso})\big)$ is inverted numerically.

At the same areal radius $R = u\,GM/c^2$, the metric components differ by

$$\frac{\Delta g_{00}}{g_{00}} = \frac{g_{00}^{\rm exp}(R) - g_{00}^{\rm Schw}(R)}{g_{00}^{\rm Schw}(R)} \sim O\!\left(\frac{GM}{c^2 R}\right)^3\,, \quad (8)$$

consistent with the $O((r_{\rm s}/r)^3)$ first discrepancy noted in Paper 2. Table **??** gives numerical values.

ISCO comparison

The innermost stable circular orbit (ISCO) is determined by the vanishing of $d^2V_{\rm eff}/dR^2$. Numerical evaluation gives:

$$R_{\rm ISCO}^{\rm exp} = 6.34\,\frac{GM}{c^2}\,, \qquad R_{\rm ISCO}^{\rm Schw} = 6.00\,\frac{GM}{c^2}\,, \quad (9)$$

a 5.7% outward shift. This modifies the final plunge waveform and the gravitational wave cutoff frequency.

Accumulated EMRI phase difference

For an extreme-mass-ratio inspiral with mass ratio $q = m/M \ll 1$, the number of orbits from radius R to $R_{\rm ISCO}$ in the adiabatic approximation is

$$N_{\rm orb}(u) = \frac{1}{64\pi q}\left(u^{5/2} - u_{\rm ISCO}^{5/2}\right), \quad (10)$$

TABLE II. Accumulated GW phase difference between exponential and Schwarzschild metrics for EMRIs, computed via Eq. (**??**). LISA phase sensitivity is $\sigma_\Phi \sim 0.1$ rad.

$M/M_\odot$	$m/M_\odot$	q	$\Delta\Phi$ (rad)	$f_{\rm ISCO}$ (mHz)
10^5	10	10^{-4}	6.7×10^3	43.8
10^6	10	10^{-5}	6.7×10^4	4.38
10^6	30	3×10^{-5}	2.2×10^4	4.38
10^7	10	10^{-6}	6.7×10^5	0.44

where $u = Rc^2/(GM)$. The accumulated phase difference between the exponential and Schwarzschild waveforms is

$$\Delta\Phi = \frac{5}{32\,q}\int_{u_{\rm ISCO}}^{u_{\rm max}} \frac{\delta\omega}{\omega}(u)\;u^{3/2}\,du\,, \quad (11)$$

where $\delta\omega/\omega$ is the fractional orbital frequency difference at areal radius $u\,GM/c^2$.

Numerical integration from $u_{\rm ISCO} = 6$ to $u_{\rm max} = 100$ gives, for representative EMRI parameters:

Prediction.—For a standard EMRI ($M = 10^6\,M_\odot$, $m = 10\,M_\odot$):

$$\boxed{\Delta\Phi \approx 6.7\times10^4\ {\rm rad}}\,. \quad (12)$$

With LISA phase sensitivity $\sigma_\Phi \sim 0.1$ rad, this corresponds to a detection significance of $\sim 7\times10^5\,\sigma$. Even if the exponential metric contributes only as a fractional correction α to the Schwarzschild geometry, LISA can detect deviations as small as

$$\alpha_{\rm min} = \frac{\sigma_\Phi}{\Delta\Phi} \approx 1.5\times10^{-6}\,. \quad (13)$$

Comparison with LQG predictions.—Zi & Kumar [**?**] found that LISA can detect loop quantum gravity modifications in EMRIs at the 2×10^{-6} fractional level. The exponential metric prediction (**??**) is $\sim 10^5$ times larger than the LQG effect, making it far easier to detect—or exclude.

Consistency with current LIGO observations

LIGO observes the final ∼10–30 orbits of comparable-mass binary mergers at $r \sim 3$–$5\,r_{\rm s}$. At these radii, $\delta\omega/\omega \sim 1$–$2\%$. Over ~ 20 orbits, the accumulated phase difference is $\Delta\Phi \sim 1$–4 rad. Current systematic uncertainties in LIGO waveform models are ~ 1 rad [**?**], so the exponential metric is *not yet excluded* by LIGO observations but lies at the boundary of detectability. The GWTC-4 analysis finding consistency with GR constrains the exponential metric admixture to $\alpha \lesssim 0.1$–1 (order-of-magnitude estimate). Next-generation detectors (ET, CE) with $\sim 10\times$ better SNR will provide definitive tests.

PREDICTION 3: GRAVITATIONAL WAVE PHASE DIFFUSION

GW decoherence from the Sigma framework

From Paper 2, the Pikovski gravitational decoherence rate in the τ framework is

$$\tau_{\rm grav}(t) = 1 - \exp\!\left(-N\,\Gamma\,t^2\right), \tag{14}$$

with $\Gamma = (\Delta E)^2\,(\Delta\Phi)^2/(2\hbar^2 c^4)$. In terms of Σ: $\Sigma_{\rm grav} = 2N\Gamma t^2$.

A gravitational wave with strain h and angular frequency $\omega_{\rm GW}$ produces a tidal acceleration $g_{\rm eff} = h\,\omega_{\rm GW}^2\,\Delta x$ across a quantum superposition of size Δx, giving a decoherence timescale

$$t_{\rm dec} = \frac{\hbar}{m\,g_{\rm eff}\,\Delta x} = \frac{\hbar}{m\,h\,\omega_{\rm GW}^2\,\Delta x^2}\,. \tag{15}$$

For a nanoparticle interference experiment (m = 170 kDa [**?**], $\Delta x = 1\ \mu$m) and a GW event ($h = 10^{-21}$, $f = 100$ Hz, $\Delta t = 0.1$ s):

$$\begin{aligned} t_{\rm dec} &\approx 9.5\times 10^{14}\ {\rm s}\,, \\ \Sigma_{\rm grav} &= 2\,\Delta t/t_{\rm dec} \approx 2.1\times 10^{-16}\,. \end{aligned} \tag{16}$$

Individual gravitational wave events produce completely negligible decoherence in any realistic quantum system.

Phase diffusion exponent

The Σ framework makes a more consequential prediction for the *propagation* decoherence of gravitational waves themselves. Cang & Wang [**?**] showed that quantum spacetime fluctuations produce phase diffusion with variance

$$\mathrm{var}(\phi) \propto d\cdot f^{\alpha_{\rm dec}}\,, \tag{17}$$

where d is the propagation distance and the frequency exponent $\alpha_{\rm dec}$ discriminates between quantum gravity models.

In the Σ framework, information loss accumulates as $\Sigma_{\rm grav} = D(\rho\|\sigma)$ along the propagation path. Since the QRE is additive (extensivity, Paper 1) and frequency-independent at leading order, the phase variance scales as d with no frequency dependence:

$$\boxed{\alpha_{\rm dec} = 0 \quad \text{(holographic scaling)}}\,. \tag{18}$$

This is distinguished from competing models:

Model	$\alpha_{\rm dec}$
String foam	-1
$\Sigma = 2\ln Q$ (this work)	0
Causal set	$+1$

Prediction.—Current bounds from Planck + BICEP/Keck + LVK + BBN [**?**] are consistent with $\alpha_{\rm dec} = 0$. The Einstein Telescope and Cosmic Explorer will improve sensitivity by $\sim 100\times$, enabling a definitive measurement of $\alpha_{\rm dec}$.

SUMMARY OF FALSIFIABLE PREDICTIONS

Table **??** collects all quantitative predictions with specific numerical values and detector requirements.

The most discriminating near-term test is the search for gravitational wave echoes at ~ 2.5 ms delay in existing LIGO–Virgo–KAGRA data using short-delay templates. The most powerful future test is the EMRI phase shift measurable by LISA, which can detect deviations from Schwarzschild at the $\alpha \sim 10^{-6}$ level.

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TABLE III. Falsifiable gravitational wave predictions of the $\Sigma = 2\ln Q$ framework. All predictions have specific numerical values testable by identified experiments.

Prediction	**Value**	**Detector**	**Sensitivity**	**Timeline**
No scalar GW mode	$h_s < 0.05\, h_t$	LIGO/ET/CE	$h \sim 10^{-23}$	Now (null result)
EMRI phase shift	$\sim 10^4$–10^6 rad	LISA	0.1 rad	2035+
GW echoes (stellar BH)	$\Delta t = 2.5$ ms	LIGO O5	Need templates	2025–2027
GW echoes (SMBH)	$\Delta t = 165$ s	LISA	Standard search	2035+
QNM frequency shift	-4.4%	ET/CE	$\sim 1\%$	2035+
Shadow size	$+4.6\%$	ngEHT	~ 2–3%	2030+
Phase diffusion exponent	$\alpha_{\rm dec} = 0$	ET/CE	$100\times$ improvement	2035+
GW friction Ξ_0	$= 1$ exactly	Standard sirens	10^{-3}	2035+
ISCO shift	$+5.7\%$	LISA (EMRI)	$\sim 1\%$	2035+